

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #08 | Digital Campus Library Reservation Gateway

COMPLETE REQUIREMENTS TABLE

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow members to search the ISBN repository and place hold requests on currently loaned books to enter a FIFO waitlist.	High	Pass: User is assigned correct position in queue. Fail: A user already holding the maximum allowed books is able to place an additional reservation.	Core reservation functionality; guarantees fair, first-come-first-served access to high-demand titles.
FR-002	Functional	The system shall notify a student member that their reserved book is available for pickup within 5 minutes of the book being checked in and marked available.	High	Pass: The member receives a notification within 5 minutes of the book's check-in timestamp for 95% of test cases. Fail: Notification is delayed beyond 5 minutes or not delivered.	Prevents queue stagnation and wasted holding-shelf space by ensuring the member acts on the hold promptly.
FR-003	Functional	The system shall automatically calculate overdue fines at a rate of Rs. 5 per day per book, capped at Rs. 200 per book, starting the day after the due date.	Medium	Pass: Calculated fine matches the formula exactly for test cases at 1, 10, 40, and 60+ days overdue (cap applied). Fail: Any test case returns an incorrect fine amount.	Ensures consistent, transparent, and auditable fine tracking without manual librarian calculation.
FR-004	Functional	The system shall allow a student member to cancel their own pending hold request at any time before the reserved copy is checked out to them.	Medium	Pass: Cancelled hold is removed from the queue immediately and the next member in line advances one position. Fail: Cancelled hold remains visible in the active queue.	Keeps the FIFO queue accurate and frees reserved copies for other waiting members as early as possible.
FR-005	Functional	The system shall allow the Head Librarian to manually reorder or remove entries in a book's holding queue for exceptional cases (e.g., a lost or damaged copy).	Low	Pass: The queue reflects the librarian's manual change and logs the librarian ID, timestamp, and reason. Fail: The change is applied without being logged, or is not applied at all.	Provides administrative flexibility to handle real-world exceptions that a purely automated FIFO queue cannot resolve.
NFR-001	Non-Functional	The catalog search query shall return paginated results for 50,000 titles in less than 150 ms.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load. Fail: p95 latency exceeds 150 ms or security checks fail under load.	Search is the entry point to every workflow; slow or insecure search degrades the experience for the entire student body.
NFR-002	Non-Functional	The system shall maintain at least 99.5% uptime during official library operating hours (8:00 AM–10:00 PM), excluding pre-announced maintenance windows.	High	Pass: Monitoring logs show uptime ≥ 99.5% over a rolling 30-day period. Fail: Uptime falls below 99.5% during operating hours in any 30-day window.	Students depend on the gateway during open hours for time-sensitive holds and fine payments; downtime directly blocks core library operations.

Use-Case Flow Specification

Digital Campus Library Reservation Gateway

Use Case ID	UC-03
Use Case Name	Manage Holding Queue
Primary Actor	Head Librarian
Supporting Actor	Digital Campus Library Reservation Gateway (System)

1. Preconditions

1. The Head Librarian is authenticated and has permission to manage holding queues.
2. The selected book exists in the library catalog.
3. The selected book has an active holding queue containing one or more member entries.
4. A valid exceptional reason exists if the Head Librarian intends to manually reorder or remove an entry.

2. Postconditions

On Success: The requested queue change is applied, the remaining entries are re-sequenced in the correct order, and the Head Librarian's ID, timestamp, and reason are recorded. Affected members are notified when applicable.

On Failure: No queue modification is applied, the existing queue remains unchanged, and no successful-change log is created.

3. Main Success Scenario

1. The Head Librarian selects a book and opens its holding queue.
2. The system displays the queue entries in their current order with relevant member and position details.
3. The Head Librarian reviews the queue and selects an entry that requires administrative action.
4. The Head Librarian chooses to reorder the selected entry or remove it from the holding queue.
5. The system requests confirmation and a reason for the manual change.
6. The Head Librarian provides the reason and confirms the change.
7. The system applies the approved modification to the holding queue.
8. The system re-sequences the remaining queue entries and updates their positions.
9. The system records the Head Librarian's ID, timestamp, action performed, and reason for the change.
10. The system identifies members whose queue positions were affected and sends the required notification.
11. The system displays the updated holding queue.
12. The use case ends successfully.

4. Alternate Flow – Head Librarian Cancels the Change

- 5a. The Head Librarian decides not to proceed with the proposed queue modification.
 - 5a1. The Head Librarian cancels the confirmation process.
 - 5a2. The system discards the pending modification.
 - 5a3. The system retains the existing holding queue without changing any positions.
 - 5a4. The use case ends without applying or logging a queue modification.